

RUBEN MARCELO BAPTISTA SILVA

EDUCATION

BSc (Licenciatura) in Computer Engineering

Polytechnic Institute of Bragança (ESTIG/IPB) | Sep 2022 – Jul 2026

- Main areas: Programming (Java, C, Python, C#), Web Development, Distributed Systems, Networks, Cybersecurity.
- Final-year project defended on 1 July 2026.

FINAL-YEAR PROJECT

Computational Study and Modelling of the LiDAR Sensor on iOS Devices

ESTIG/IPB | 2025 – 2026

- Developed a full iOS 3D-scanning application (Swift, SwiftUI, ARKit) with three capture modes: environment mesh (OBJ), filtered object point cloud (CSV/PLY) and photogrammetry dataset capture.
- Built a Python analysis pipeline (NumPy, SciPy, Matplotlib) and derived an empirical power-law noise model for the Apple LiDAR sensor ($R^2 = 0.97$).
- Performed photogrammetric reconstruction of captured datasets in Meshroom (CUDA/GPU).

PUBLICATIONS

- Scientific paper on LiDAR sensor characterisation and 3D scanning on iOS — submitted to the **TEEM 2026** international conference (under review).
- Scientific paper derived from the same research line — submitted to the **ARTIIS 2026** international conference (under review).

EXPERIENCE

BIP — Vilnius College, Design & Technical Faculty

Vilnius, Lithuania | May 2025

- Practical robotics project using a robotic arm, integrating software and hardware.
- Developed team communications and troubleshoot system faults.

Erasmus Programme — University of Split

Split, Croatia | Feb 2024 – Jul 2024

- One-semester international exchange, improving autonomy, adaptability and teamwork.

TECHNICAL SKILLS

- **Mobile / AR:** Swift, SwiftUI, ARKit
- **Programming:** Python (NumPy, SciPy, Matplotlib), Java, C/C++, C#, JavaScript
- **Web:** HTML, CSS, React (basic)
- **Tools:** Git/GitLab, LaTeX, Xcode, Linux, Docker (basic)
- **Systems:** Virtualisation and self-hosting (Proxmox, VMware Workstation), networking fundamentals

CONTACT

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Nationality: Portuguese

SUMMARY

Computer Engineering graduate (July 2026) from the Polytechnic Institute of Bragança. Author of two scientific papers on LiDAR sensor modelling, developed from an iOS 3D-scanning application built as a final-year project. International experience through Erasmus (Croatia) and a BIP programme (Lithuania). Motivated, responsible and adaptable, seeking to deepen technical expertise through an MSc in Informatics Engineering.

LANGUAGES

Portuguese — Native

English — C1 (Advanced)